

WEEK 7

Week 7: Parallel Algorithms - Lab Guide

HPC & Cluster Computing

Objective: designing a decomposition strategy for a 2D problem on paper (or whiteboard).

Part 1: Domain Decomposition

Scenario: You have a 16×16 grid representing a metal plate. You have 4 MPI Processes.

1. **Block Decomposition**: Draw how you would split the grid if you gave each rank a 8×8 block.
2. **Strip Decomposition**: Draw how you would split the grid if you gave each rank 4 full rows (4×16).
3. **Ghost Cells**:
   - Select Rank 1.
   - Identify which rows/columns belong to it.
   - Draw the "Ghost Cells" it needs from its neighbors.
   - How many integers does it s[illegible]ny does it receive?

Part 2: Functional Decomposition

Scenario: You are building a real-time video processing pipeline for a security camera.

Stages:

1. **Decode**: Read raw bits. (Heavy CPU)
2. **Filter**: Enhance night vision. (Moderate CPU)
3. **Analyze**: Face recognition AI. (Very Heavy CPU / GPU)
4. **Save**: Write to disk. (I/O Bound)

Task:

- You have 10 nodes.
- How would you assign them to these 4 stages? (e.g., 2 nodes for Decode, etc.) to minimize bottlenecks?
- Justify your choice. (Hint: The "Analyze" stage is the slowest).

